

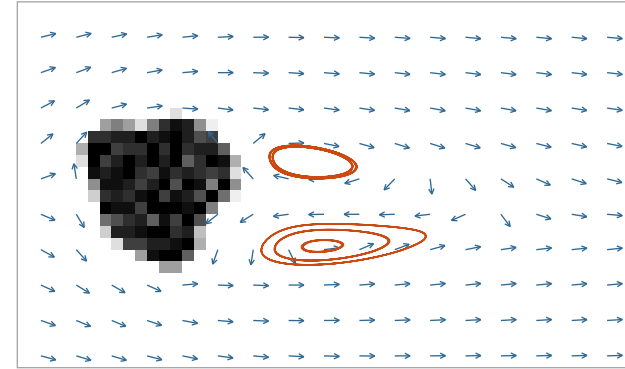
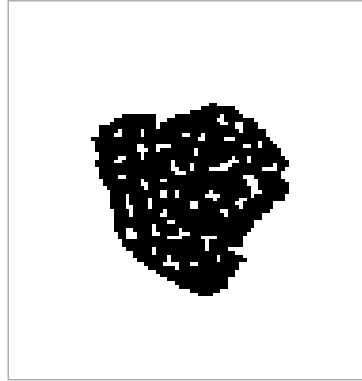
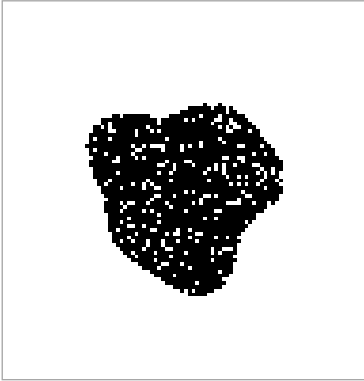
# Binary material can maintain an obstruction to flow

Prepared material

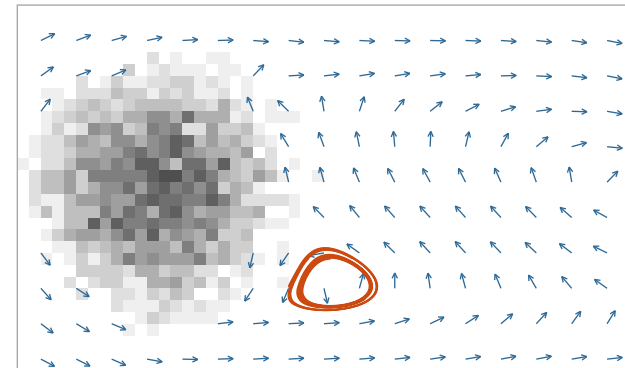
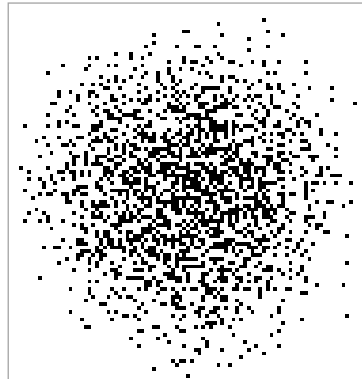
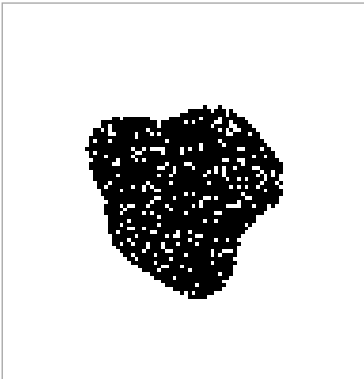
After 140 local sweeps

Flow: final snapshot

With cohesion



Without cohesion



Same starting material in both rows. Blue: velocity direction. Orange: selected returning paths in the frozen final field.

The substrate supports the material. These runs test persistence from a prepared patch, not spontaneous formation from uniform noise.